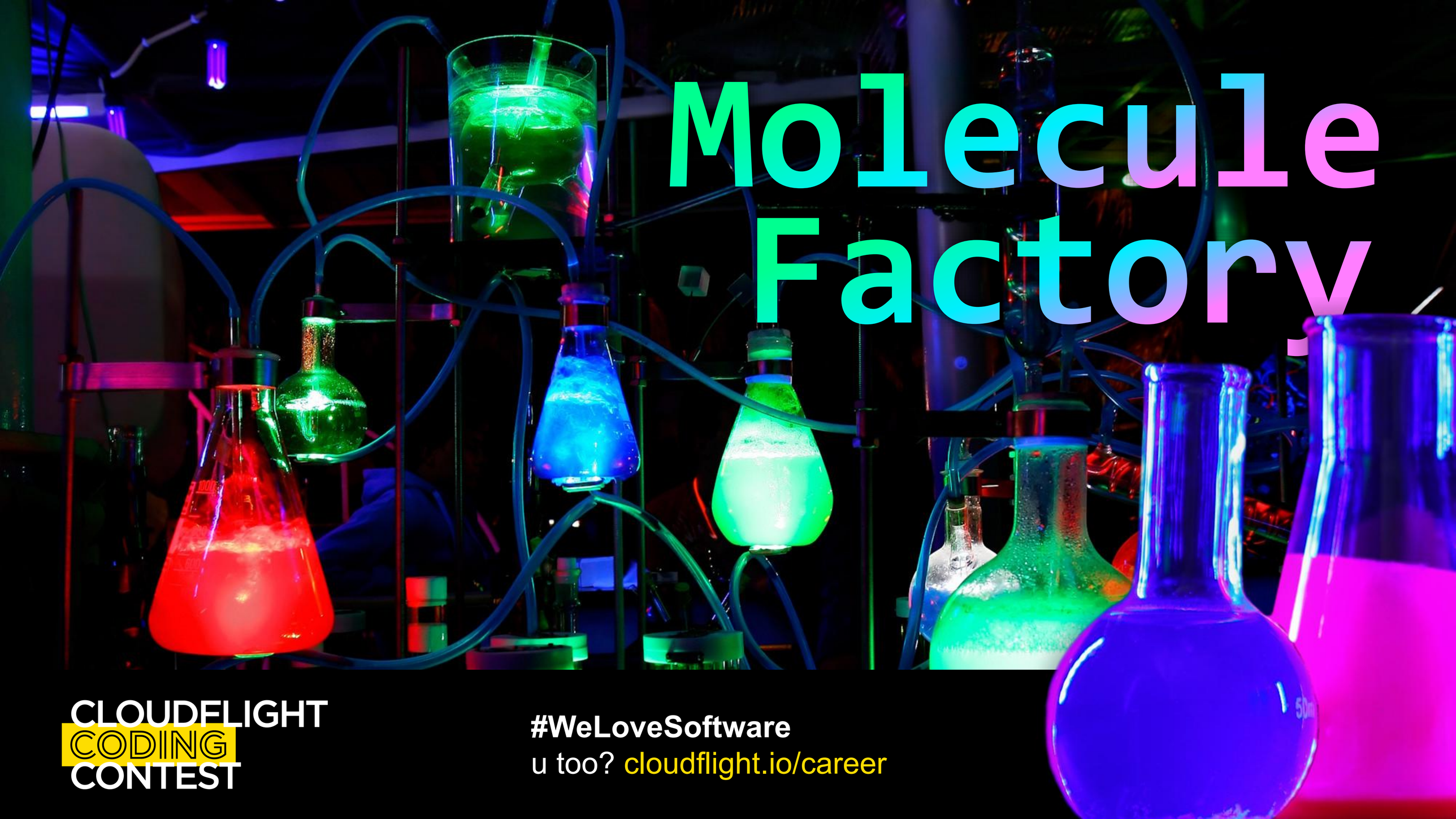


CLOUDFLIGHT
CODING
CONTEST

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WELCOME TO THE LAB

Behind an unmarked door, under a blacklight, Professor Flux has built the world's most ridiculous chemistry lab: kilometers of glowing glass tubing, bubbling flasks, and a dishwasher that hasn't washed a dish since 2009.

The machine can turn air, water and a pinch of charcoal into molecules no one has ever seen before.

There's just one problem: the Professor has no patience for paperwork. The factory does exactly what it's told, nothing more and nothing less.

That's where you come in.

You are here to program the factory, just like you do with your computer.

Level 1



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In the factory, molecules are transformed with chemical reactions to produce new molecules.

- Reactions transform molecules into other molecules
- Reactions can have multiple input molecules and multiple output molecules

Each input file has multiple cases, for each:

- You are given an inventory of molecules
- You are given a reaction
- **Perform the reaction once and print the resulting inventory**



Input

Name		Description	Example
N		Number of cases	3 3 H2, 4 O2, 7 H2O 2 H2, 1 O2 -> 2 H2O 6 CO2, 7 H2O 6 CO2, 6 H2O -> 1 C6H12O6, 6 O2 3 C6H12O6 1 C6H12O6 -> 2 C2H5OH, 2 CO2
repeated N times	Inventory	**A molecule list** A list of molecules separated by comma ‘,’ Each molecule is a molecule count and a molecule name separated by a space ‘ ’ *A molecule name can be any alphanumeric string, and it will be used only as an identifier for the rest of the contest*	
	Reaction	Two **molecule lists** separated by an arrow “ -> ”	
	Empty line between cases		

Output

Name	Description	Example
Resulting inventory (repeated N times)	**The resulting molecule list**	1 H2, 3 O2, 9 H2O 1 H2O, 1 C6H12O6, 6 O2 2 C6H12O6, 2 C2H5OH, 2 CO2

For each input file, determine the output and write it in a file.
Upload your output file to the coding contest platform.



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